Baseline Generation Gallery and Memo

IE 7615 Deep Learning for AI · Project 2, Milestone 1 · Group 02: Abdellah Faleh, Jin-Woo Hong, Samuel Tong, Tristan Lyons · October 2026

What we ran

We gave 15 captions from our **test set** to a **pretrained Stable Diffusion 1.5** model, without any training. We picked about two captions for each scene type (animals, sports, water, city, nature, indoor, people). Settings: 30 denoising steps (PNDM scheduler), guidance scale 7.5, 512×512 pixels, and a fixed seed for each caption (42 + caption number), so every image can be made again. Each image took about 4.9 seconds on a free Colab T4 GPU.

To put numbers on the comparison we used **CLIP ViT-L/14**. The **CLIP score** (0 to 100) shows how well an image matches its caption; higher is better.

Measure	Result
Average CLIP score, generated images	27.0
Average CLIP score, real Flickr30k photos (same captions)	24.4
Captions where the generated image scores higher than the real photo	12 of 15
Average CLIP similarity, generated image vs. real photo	71.9
Images blocked by the safety filter	0 of 15

Table 1. Summary scores for the 15 test captions.

Diversity test (same caption, four seeds)

To check for **mode collapse**, we made 4 images for each of 3 captions with seeds 1000 to 1003, and measured how similar the 4 images are to each other (CLIP similarity, 0 to 100; lower means more variety).

Caption	Similarity between the 4 images
A tan dog running through the grass with the background blurry	93.0
A dog with golden fur is chest deep in water	88.1
A man in a number ten jersey kicks a soccer ball while another team member watches	81.8

Table 2. Diversity across seeds.

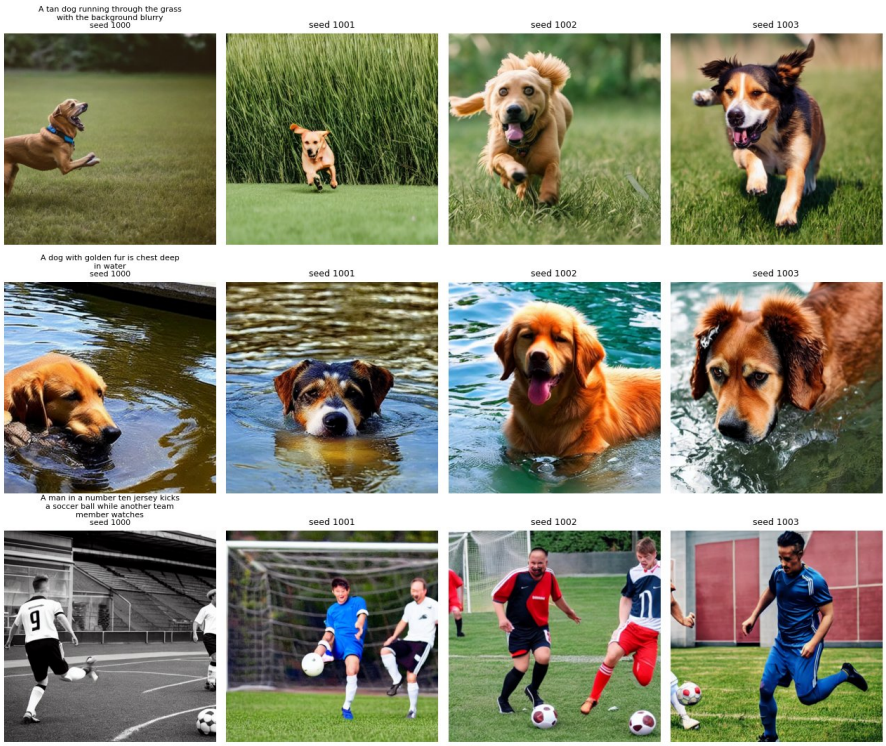


Figure 1. Diversity test: each row is one caption; each column is a different seed.

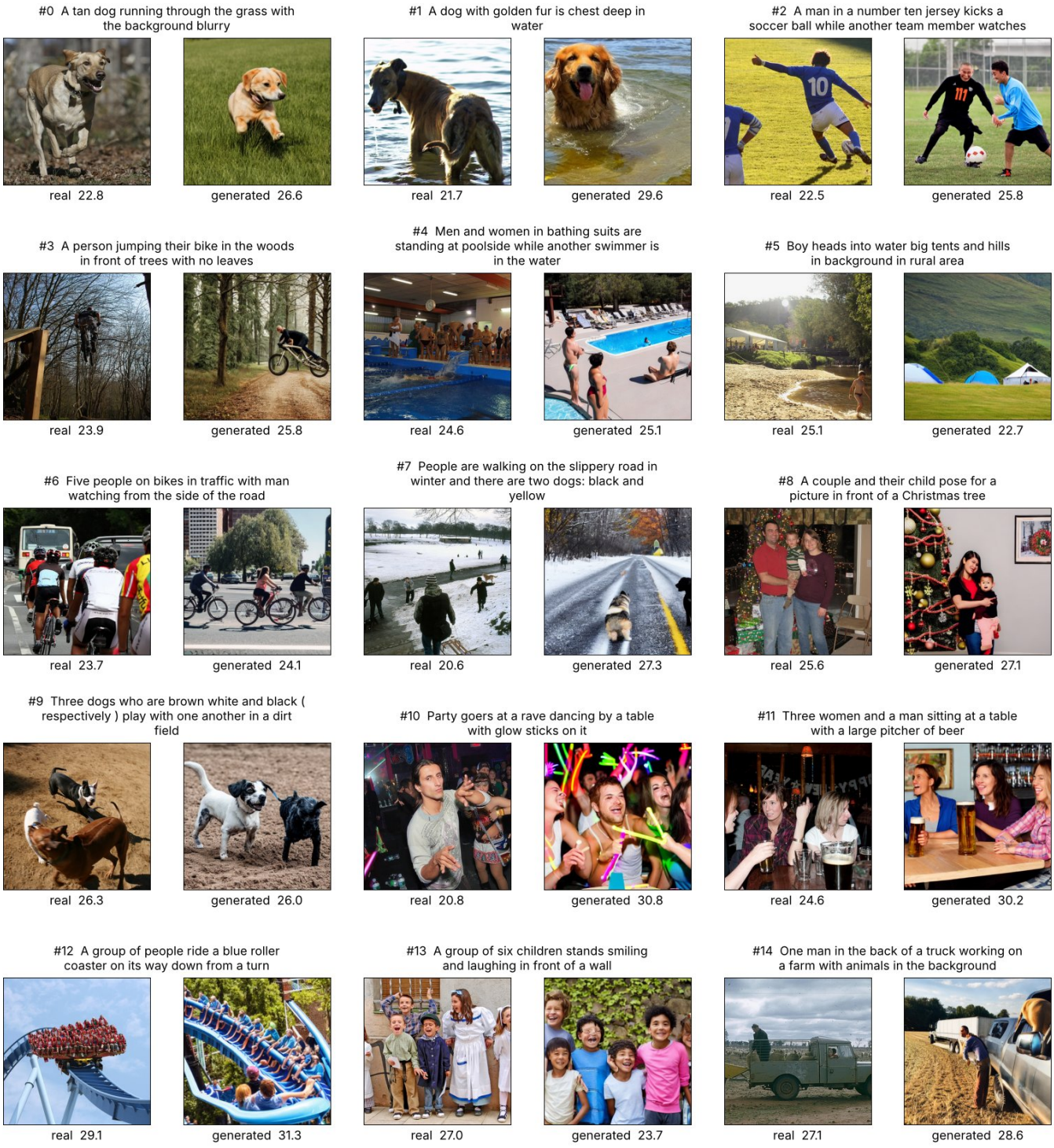


Figure 2. All 15 captions: real Flickr30k photo (left) and generated image (right). The numbers are CLIP scores.

Qualitative analysis

Overall, the images follow the captions well. For 12 of 15 captions the generated image has a higher CLIP score than the real photo. The model draws exactly what the caption says, clearly and in the center, while real photos have extra objects, strange angles and bad light. Note that CLIP score uses the same kind of model that guides Stable Diffusion, so it may favor generated images. FID in Milestone 3 will check how *real* they look.

What worked well: animals (#0, #1, #9), and scenes with one strong idea: the roller coaster (#12), the rave with glow sticks (#10) and the family by the Christmas tree (#8).

Typical mistakes:

- **Counting:** "five people on bikes" gave three (#6); "three dogs" gave two (#9); "three women and a man" gave three women and no man (#11).
- **Numbers and text:** "number ten jersey" shows "111" (#2). In Figure 1 the jerseys show other numbers (9, 11), never 10.
- **Missing subjects in long captions:** "boy heads into water" shows only tents and hills (#5), the lowest score (22.7); the people on the snowy road are missing in #7.
- **Negation:** "trees with no leaves" gave a forest with leaves (#3). The text encoder sees the word "leaves" and draws them.
- **Bodies and style:** some bodies at the pool look wrong (#4), and all images look brighter and cleaner than Flickr photos, like stock photos (similarity to the real photo only 71.9).

Generative vs. discriminative framing. In Project 1 a discriminative model learned p(label | image): one correct answer per photo. Here the model learns p(image | text): one caption fits many different pictures. So a generated image can be correct and still look nothing like the real photo (compare the two roller coasters in #12). That is why we judge it with CLIP score and FID, not by comparing pixels with one real photo.

Latent representation. Stable Diffusion does not remove noise from the full picture. It works on a compressed *latent* of 64×64×4 numbers, 48 times smaller than the 512×512 color image, and a decoder turns it into the picture at the end. This is why one image takes only about 5 seconds on a free GPU. The small latent may also explain the weak small details: faces in crowds, hands and jersey numbers.

Conditional sampling. Each image is one random sample from the model "given" the caption. The caption guides every denoising step through cross-attention, and the guidance scale (7.5) sets how strongly. The seed decides everything else, which is why Figure 1 shows four different scenes for the same caption.

Mode collapse risk. Mode collapse means a model keeps making nearly the same image. We do not see it: the four seeds give different poses, backgrounds and colors. But variety depends on the caption. The dog caption is the least varied (93.0, always a tan dog on green grass), while the soccer caption is the most varied (81.8, even a black-and-white photo). A high guidance scale can reduce variety, so in Milestone 2 we will test guidance scales from 1 to 15.

Next steps. Milestone 2 will feed our own CLIP embeddings into the model and compare CLIP with BERT, test the guidance scale, scheduler and number of steps, and check whether these changes fix the counting and missing-subject mistakes seen here.